

Lu Wang

PHD STUDENT

✉ lwang97@stevens.edu | 🏠 luwang823.github.io | 🌐 LuWang823 | in LuWang823

Summary

I am a Human-Computer Interaction and Health Informatics researcher, applying social science knowledge and methods to inform design and development. My focus is on applying Artificial Intelligence (AI) in healthcare, including users' needs investigation, unobtrusive design and development, and privacy and fairness issues.

My Ph.D. thesis investigates how to support informal caregivers through conversational agents and how current generative AI can be designed to promote effective and safe use.

Education

PhD Student in Computer Science, Transferred

Hoboken, NJ, US

STEVENS INSTITUTE OF TECHNOLOGY

2024-2027

- GPA: 3.80/4.00
- Advisor: Dr. Jina Huh-Yoo
- Research Areas: Human-Computer Interaction, Health Informatics, Artificial Intelligence

PhD Candidate in Information Science, College of Computing & Informatics

Philadelphia, PA, US

DREXEL UNIVERSITY

2020-2024

- GPA: 4.00/4.00
- Advisor: Dr. Jina Huh-Yoo
- Research Areas: Human-Computer Interaction, Health Informatics, Artificial Intelligence

Master of Applied Psychology, User Experience

Beijing, CN

BEIJING NORMAL UNIVERSITY (BNU)

2016-2018

- GPA: 87/100, GPA Rank: 1/59
- Advisor: Dr. Jian Li
- Thesis: Using Multiple Source Information to Predict User's Emotion
- Keywords: User experience, Emotion prediction, Text mining, Machine learning, Chat-log

Bachelor of Psychology

Beijing, CN

BEIJING NORMAL UNIVERSITY (BNU)

2012-2016

- GPA: 90/100
- Advisor: Dr. Jian Li
- Thesis: Recovery of Resources is for Better Learning: The Influence of College Students' Daily Learning on Their Self-control Resources
- Keywords: Self-depletion, Diary study, Hierarchical linear model, Dumbbell straight method

Publications and Conferences

Findings of the Association for Computational Linguistics: ACL

Vienna, Austria

RESEARCH ARTICLE

2025

- Elham Aghakhani, Lu Wang, Karla T. Washington, George Demiris, Jina Huh-Yoo, and Rezvaneh Rezapour. 2025. From Conversation to Automation: Leveraging LLMs for Problem-Solving Therapy Analysis. In Findings of the Association for Computational Linguistics: ACL 2025, pages 25189–25207, Vienna, Austria. Association for Computational Linguistics

Proceedings of the ACM on Human-Computer Interaction, Volume 8, Issue CSCW1

San José, Costa Rica

RESEARCH ARTICLE

November 9-13, 2024

- Diva Smriti, Lu Wang, and Jina Huh-Yoo. 2024. Emotion Work in Caregiving: The Role of Technology to Support Informal Caregivers of Persons Living With Dementia. Proc. ACM Hum.-Comput. Interact. 8, CSCW1, Article 48 (April 2024), 34 pages. <https://doi.org/10.1145/3637325>

Social Psychology

Journal

RESEARCH ARTICLE

June 16, 2020

- Wen, Wen, Jian Li, George Kyriacou Georgiou, Cheng Huang, and Lu Wang. 2020. Reducing the halo effect by stimulating analytic thinking. Social Psychology

ACM CHI Conference on Human Factors in Computing Systems

Honolulu, Hawai'i

LATE-BREAKING WORK

May 11-16, 2024

- Lu Wang, Diva Smriti, Hao Yuan, and Jina Huh-Yoo. 2024. Artificial Intelligence Systems for Supporting Informal Caregivers of People Living with Alzheimer's Disease or Related Dementias: A Systematic Review. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 48, 1–11. <https://doi.org/10.1145/3613905.3650846>

ACM CHI Conference on Human Factors in Computing Systems

Virtually

LATE-BREAKING WORK

April 23-28, 2023

- Lu Wang, Chaomei Chen, and Jina Huh-Yoo. 2023. Investigating the Synonyms of Conversational Agents to Aid Cross-Disciplinary CA Research. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 190, 1–10. <https://doi.org/10.1145/3544549.3585640>

Proceedings of the Association for Information Science and Technology

Washington, D.C.

POSTER, PRESENTER: DUNCAN LYNCH

Nov. 14-18, 2025

- Lu Wang, Duncan Lynch, Elham Aghakhani, George Demiris, Karla Washington, Rezvaneh Rezapour, and Jina Huh-Yoo. 2025. AnnoLoom: Augmenting Codebook Generation and Annotation with Large Language Models. Proceedings of the Association for Information Science and Technology 62, 1 (October 2025), 1711–1713. <https://doi.org/10.1002/pra2.1517>

AMIA 2025 Annual Symposium

Atlanta, GA

POSTER

Nov. 15-19, 2025

- Ananya Shrivastava, Lu Wang, Elham Aghakhani, Rezvaneh Rezapour, and Jina Huh-Yoo. 2025. Enhancing Transparency in Large Language Model-Based Research with PromptLog: A Tool for Prompt History Tracking and Reporting. In AMIA Annual Symposium Proceedings 2025. American Medical Informatics Association.

Computing Research Association Widening Participation Grad Cohort for Women

San Francisco, CA

POSTER

April 20-22, 2023

- Lu Wang and Jina Huh-Yoo. 2023. Validity Examinations of Social Bias Measurements and Mitigations in Word Embeddings

AMIA 2021 Annual Symposium

Hilton San Diego Bayfront

POSTER

Oct. 30 2021 - Nov. 3 2021

- Lu Wang and Jina Huh-Yoo. 2021. The Representativeness of “People Also Ask” of Google Web Search on the Information Needs Concerning Alzheimer's Disease and Related Dementias. In AMIA Annual Symposium Proceedings 2021. American Medical Informatics Association.

Computing Research Association Widening Participation Grad Cohort for Women

Virtual

TECH TALK

Apr. 2021

- Lu Wang, Munif Mujib, Jake Williams, George Demiris, and Jina Huh-Yoo. 2021. An Evaluation of Generative Pre-Training Model-based Chatbot in Therapist-Patient Dialogue Context

Annual Conference on Digital Media Studies (DMS'18)

Beijing, China

TALK

Nov. 2018

- Lu Wang and Jian Li. 2018. Using Multiple Source Information to Predict User's Emotion

The International Association of Societies of Design Research (IASDR'17)

Cincinnati, US

TALK

Nov. 2017

- Zhiyong Fu, Jia Lin, and Lu Wang. 2017. The Research and Design Framework for Citizen Science

Industrial Design Research (UXPA China'17)

Nanjing, China

TALK

Oct. 2017

- Ying Fu, Han Xu, Lu Wang, Mengfan Li, and Yu Dai. 2017. ZANMA-Improve the Relationship between Mother-in-Law and Daughter-in-Law

31st International Congress of Psychology

Yokohama, Japan

POSTER

July 24–29, 2016

- Lu Wang, Wen Wen, Jian Li, and Cheng Huang. 2016. Cool the Warm: Stimulating Analytical Thinking to Make Rational Decisions

arXiv

PRE-PRINT

- Wang, L., Song, M., Rezapour, R., Kwon, B. C., & Huh-Yoo, J. (2023). People's perceptions toward bias and related concepts in large language models: A systematic review. arXiv preprint arXiv:2309.14504.

- Wang, L., Mujib, M. I., Williams, J., Demiris, G., & Huh-Yoo, J. (2021). An evaluation of generative pre-training model-based therapy chatbot for caregivers. arXiv preprint arXiv:2107.13115.

Honors & Awards

2024	Exceptional Academic Achievement Award at Department of Information Science	CCI, Drexel University
2023	Grace Hopper Celebration (GHC) Student Scholarship from Drexel University	Orlando, FL
2023	CRA-WP Grad Cohort Conference Grant	San Francisco, CA
2022	Grace Hopper Celebration (GHC) Student Scholarship	Virtual
2022	NSF SCH PI Workshop Student Award	Virtual
2021	Grace Hopper Celebration (GHC) Student Scholarship	Virtual
2021	CRA-WP Grad Cohort Conference Grant	Virtual
2021	NSF SCH PI Workshop Student Award	Virtual
2021	GREPSEC Workshop Student Award	Virtual
2021	USENIX Diversity Grant	Virtual
2018	Outstanding Graduates (Top 5%)	Beijing
2018	University Houcan Scholarship	School of Psychology, BNU
2017	First Class Scholarship (Top 10%)	Beijing Normal University
2017	Huawei Scholarship	Huawei Technologies Co., Ltd.
2016	Outstanding Graduates (Top 5%)	Beijing
2016	Beisen Scholarship	Beijing Beisen Cloud Computing Co.,Ltd.
2015	National Endeavor Fellowship	China
2014	National Endeavor Fellowship	China
2014	First Class Scholarship (Top 10%)	Beijing Normal University
2014	Second Prize of Mathematical Modeling Competition	Beijing Normal University
2013	National Endeavor Fellowship	China
2013	First Class Scholarship (Top 10%)	Beijing Normal University
2013	Third Prize of ACM Computer Programming Competition	Beijing Normal University

Skills

Psychology Experiment	E-Prime, Inquisit, Qualtrics
Programming	Python, Javascript, MATLAB, C, Pascal
Statistics	R, SPSS Statistics, Mplus
Design	Adobe Photoshop, Adobe Illustrator, Flash
Research Methods	Quantitative Methods, Qualitative Methods, Participatory Design Methods, Statistical Visualization

Work Experience

Research Assistant	Hoboken, NJ, US
STEVENS INSTITUTE OF TECHNOLOGY	August 2024-present
- Performed systematic literature reviews, analyzing over 150 papers on prompt intervention and behaviors of Large Language Models(LLMs). - Conducted interviews to understand how Generative Conversational AI can be applied and designed to support informal caregivers. - Compared results of “People Also Ask” of Google Web Search and multiple LLMs regarding questions on dementia and Alzheimer’s. - Designed a Chrome Extension to track and report prompt history for researchers. - Designed a LLM-supported annotation platform to assist annotators in developing codebooks, iteratively prompting LLMs for auto-annotation, systematically comparing annotation results between LLMs and human annotators, and reusing the developed codebooks and prompts for real-time annotation and sharing across researchers.	

Research Assistant

Philadelphia, PA, US

DREXEL UNIVERSITY

May 2020-July 2024

- Performed literature reviews, analyzing over 200 papers across various topics such as conversational agents, AI support for caregivers of people living with dementia, and people's perceptions toward bias and related concepts in large language models.
- Conducted mixed methods (quantitative sentiment analysis and qualitative content analysis) to evaluate the GPT-2 and fine-tuned GPT-2 performances for Problem-Solving Therapy based on 306 therapy session transcripts between informal caregivers of people living with dementia and therapists. Coded and analyzed interviews of 12 informal caregivers of people living with dementia to understand when technology should or should not support the emotional work engaged in informal caregiving.
- Applied web crawl methods to investigate the representativeness of "People Also Ask" of Google Web search on the information needs concerning Alzheimer's Disease and related dementias and compare ChatGPT with users when responding to Reddit posts related to Alzheimer's Disease and related dementias.

User Researcher

Hangzhou, China

NETEASE INC. (NASDAQ: NTES)

July 2018 - Dec. 2018

- Conducted user studies for the business line of NetEase Cloud Music
- Completed seven projects from the early planning stage of the product to the iteration of the NetEase Cloud Music App. Working methods included the interview method, questionnaire method, web crawler, persona, etc.

Data Analyst Intern

Beijing, China

GXHS TECHNOLOGY

Nov. 2017 - Jan. 2018

- Assisted in paper reviewing, data analysis, data visualization, and report writing.
- Accomplished descriptive statistics analysis and text analysis of cases on National 12358 Price Supervision Platform.
- Applied R language to analyze Baidu Index, base station data, and other indicators to describe the flow of Xiong'an talents.
- Analyzed and compared the emotional index differences of people from different media sources with a data analysis platform.

User Researcher Intern

Beijing, China

LENOVO RESEARCH

July 2017 - Oct. 2017

- Assisted in Artificial Intelligence research.
- Used R language and Python to analyze customer service dialogue data;
- Labeled the talking techniques used by customers and agents through chat logs to find the most appropriate responses to different user states and assisted in the design of the Chatbot interaction flow chart.

Teaching Assistant

Beijing, China

BEIJING NORMAL UNIVERSITY

2018, 2017, 2016

- Educational Psychology, Sept. 2016 - Feb. 2017, Assisted instructor (Dr. Jian Li) in attendance checking, instructing, and performance rating.
- Issues in Research Methodology: Quantitative, Qualitative, and Action Research, Apr. 2018 - May 2018, Assisted instructor (Dr. Xianjun Zheng) in attendance checking, group forming and instructing, performance rating, and rewarding.

User Researcher Intern

Beijing, China

BAIDU.COM TIMES TECHNOLOGY (BEIJING) CO. LTD.

Oct. 2015 - Mar. 2016

- Assisted in telephone interview invitation, interview recording, questionnaire inputting, data analyzing, and report writing.
- Developed a program to generate syntax of SPSS using C language, which improved the efficiency of data analysis by 200%.

Projects

Understanding and Anchoring People's Mental Models of Large Language Models for Problem-Solving in Informal Caregiving Context

Stevens Institute of Technology

GRADUATE RESEARCH ASSISTANT

August. 2024 — Now

- Conduct a systematic review to understand how participants interact with Large Language Models
- Conduct an interview study to understand caregivers' needs in supporting people with chronic conditions

Co-Designing and Understanding the Roles of Conversational Artificial Intelligent Systems on Caregiving

Drexel University

GRADUATE RESEARCH ASSISTANT

Apr. 2022 — July. 2024

- Conduct a review mapping study to understand the caregivers' needs and find opportunities for developing assistive tools
- Conduct a qualitative analysis of interviewing caregivers' needs in supporting people living with dementia

Assistive Tool for Conversation of Patients, Caregivers, and Doctors project

Drexel University

GRADUATE RESEARCH ASSISTANT

Mar. 2022 — May. 2022

- Implementing Microsoft Azure to realize speech-to-text function
- Outcome: Python codes

Fairness in Word Embeddings

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

Jan. 2022 — Now

- Investigate the gaps in fairness definitions, evaluations, and mitigations in word embeddings

Privacy Agreement Settings of Chatbot Development Frameworks and Their Deployment Platforms

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

Apr. 2021 — Now

- Investigate how the privacy agreement is written and designed for different chatbot development frameworks and their deployment platforms for third-party entities.

How Chatbot Concepts Evolve in Multi-disciplinary Context

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

Oct. 2020 — Now

- Use the CiteSpace software to track how chatbot concepts evolve in a multi-disciplinary context to understand how researchers from different backgrounds or disciplines understand/define these concepts.

Internet Search Analytics of Dementia-Related Questions Project

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

Dec. 2020 — Sept. 2021

- Investigate what questions people ask about Alzheimer's disease and related dementias. Used Google's "People Also Ask" function to investigate the most frequently asked questions on the Web and their suggested answers.

Evaluation of GPT-2 based Chatbot in Therapist-Patient Dialogue Context

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

July 2020 — Dec. 2020

- Evaluated the performance of the pre-trained model and the fine-tuned model of OpenAI GPT-2 from the perspective of conversational quality in the Therapist-Patient Dialogue Context. Evaluation methods included analyses of the proportion of non-word outputs, the length of response (number of words in a response), and sentiment components.

CADENCE (a Conversational Agent [CA] based library for DEmeNtia Care) Project

GRADUATE RESEARCH ASSISTANT

[Drexel University](#)

May 2020 — July 2020

- Participated in the review work of automatic summarization and analysis of web searching results related to dementia.

Assessment and Training of Metacognition Regulation Skills of Adolescents Project (The 12th Five Year Plan of Education Science)

GRADUATE RESEARCH ASSISTANT

[Beijing](#)

June 2015 — May 2018

- Participated in review work of metacognition and problem-solving, research conduction in high school, and data analysis of scales using SPSS, Mplus, and R Language.

Learning Quality Scale Revision Project (Research and Development of the Evaluation Model of the Comprehensive Psychological Quality of Primary and Secondary School Students)

GRADUATE RESEARCH ASSISTANT

[Beijing](#)

Mar. 2016 — Dec. 2017

- Participated in review work of learning quality, research conduction in high school, and items analysis and scale revision of eight scales using SPSS, Mplus, and R Language.

China Maker Education Blue Paper 2016 Project

GRADUATE RESEARCH ASSISTANT

[Tsinghua University & Intel](#)

Jan. 2017 — Aug. 2017

- Leader of the research group, mainly charged with desk research, research design, and report writing, accomplished questionnaire design and publishing, interview design, and conduction, as well as data analysis and presentation.

Driving Gamification for Older People

GRADUATE RESEARCH ASSISTANT

[Swinburne University of Technology](#)

Mar. 2017 — May 2017

- Conducted scoping review through identifying 46632 records, screening of 41549 records, eligibility examination of 133 records, and paper review of 90 records.

Cortana & Internet of Things (IoT) Project

GRADUATE RESEARCH ASSISTANT

[Microsoft Research Asia \(MSRA\)](#)

Feb. 2017 — June 2017

- Designed a smart kitchen assistant to help working women with the daily menu and fasten the relationships among families based on IoT and artificial intelligence technology (Cortana). Participated in desk research, questionnaire design, interview, persona making, journey map development, conceptual design, and presentation delivery. (Confidentiality Agreement Signed)

Natural Voice Interaction in Chinese Context Project

Land Rover

GRADUATE RESEARCH ASSISTANT

Sept. 2016 — Dec. 2016

- Designed a tool to solve fatigue driving problems. Participated in desk research, collage making, mind map making, context research, user journey map making, affinity diagram making, logic map making, toolkit research, storyboard making, low-fidelity prototyping, and iterating. (Confidentiality Agreement Signed)

Undergraduate Researcher Supporting Research Projects

Beijing

Sept. 2012 — June. 2016

- Completed paper review of Halo Effect, dual-system theory, and ego-depletion and participated in research design, research conduction of ten pilot studies, and two formal studies using scales, E-prime experiments, dumbbells, and dictation task, etc., analyzed data using SPSS.
- Conducted questionnaire survey related to the influence of parenting style on children's procrastination behavior in Chengdu City, Sichuan Province, China (Summer, 2014) and participated in the survey on sex education of eight high schools in Henan Province, China (Summer, 2015).
- Transcribed sound records of interviewing parents in mainland and Taiwan, distributed.
- Collected questionnaires and scales related to teachers' ethics and teaching evaluation system
- Conducted experiments to study attention, decision making under risks and Electroencephalogram changes when subjects played games
- Conducted interviewer studies to investigate the relationship between roommates in the dormitory and investors' thinking progress in illegal fund-raising projects

Exchange Experience

Swinburne University of Technology

Australia

VISITING SCHOLAR

Mar. 2017 - May 2017

- Projects on driving simulators for old drivers.

Professional Services and Volunteer Experience

2026	PC for CSCW 2026	<i>Virtual</i>
2025	Associate Chair for CSCW 2025 and CHI 2025 Late-Breaking Work, PC of AIES	<i>Virtual</i>
21-26	Reviewer for AMIA, CHI, CSCW, HRI, DIS, IUI	<i>Virtual</i>
2025	SV at ACM CHI conference on Human Factors in Computing Systems	<i>Yokohama, Japan</i>
2024	SV at ACM CHI conference on Human Factors in Computing Systems	<i>Hawai'i, USA</i>
2020	IEEE BigData Conference	<i>Virtual</i>
2020	Research Data Alliance (RDA) US	<i>Virtual</i>
2018	Language Rehabilitation of Hearing and Impaired Children Program	<i>Beijing, China</i>
2016	Conference of Chinese Psychologists	<i>Beijing, China</i>
2016	Chinese Applied Psychology Conference	<i>BNU, Beijing, China</i>
2015	Slush International Conference	<i>Beijing, China</i>
2014	Asia-Pacific Economic Cooperation (APEC)	<i>Beijing, China</i>
2014	Voluntary support education in Sichuan Province	<i>Jiayang, China</i>
2013	China (Beijing) Garden Expo	<i>Beijing, China</i>
2013	China Tennis Cheerleading Service for China Open	<i>Beijing, China</i>
2012	United the Force for Good Program, China Foundation for Poverty Alleviation	<i>Beijing, China</i>